Lesson 2

Material

https://github.com/unixfile/learning-python-2026

Today

- Numbers and arithmetic
- Booleans and truthiness
- Reading and writing files
- Code blocks and indentation

1 Integers and floats

```
In : foo = 42
In : type(foo)
Out: int

In : bar = 2.71828
In : type(bar)
Out: float
```

2 Basic arithmetic

```
In : result = 1 + 2 - 3 * 4
In : result, type(result)
Out: (-9, int)

In : quotient = 3 / 7
In : quotient, type(quotient)
Out: (0.4285714285714285, float)
```

3 Power and modulo

```
In : 2 ** 5
Out: 32

In : 33 % 10
Out: 3
```

4 Augmented assignment

```
n += 1    # n = n + 1
q /= 3    # q = q / 3
r %= 2    # r = r % 2

# i++ has no meaning in Python
```

5 Booleans

```
In : to_be = True
In : not_to_be = False

In : to_be or not_to_be
Out: True

In : to_be and not_to_be
Out: False
```

6 Truthiness

```
In : bool(0)
Out: False
In : bool(1600)
Out: True

In : bool("")
Out: False
In : bool("Hamlet")
Out: True

In : bool(0.0)
Out: False
In : bool(0.0001)
Out: True
```

7 Using booleans and truthiness

```
if True:
    print("Always")
if False:
    print("Never")

if 1337:
    print("Always")
if 0:
    print("Never")
```

8 Reading from a file

```
f = open("foo.txt")
content = f.read()
f.close()
type(content)  # str
```

9 Reading lines

```
f = open("bar.txt")
lines = f.readlines()
f.close()
type(lines)  # list
```

10 Iterating over lines

```
f = open("baz.txt")
for line in f:
    print(line)
f.close()
```

11 Writing to a file

```
f = open("file.txt", "w")
f.write("Travel far, ")
f.write("meet yourself.\n")
f.flush()
f.close()
```

12 Appending to a file

```
f = open("file.txt", "a")
f.write("Appending...\n")
f.flush()
f.close()
```

13 Opening with scope

```
with open("file.txt", "r") as f:
    data = f.read()
```

- File is closed automatically at end of block

14 Code blocks

Blocks start with `:` and indent 4 spaces

```
if True:
    print("banan")
    print("melon")
    for i in range(10):
        print(1 << i)
    print("kiwi")
    print("citron")
```

- Blocks can be nested